```
    M ←
[]
#4
    for i in {1,..,runsPerK} do
         #5
      (centroids, clusters) ← kmeans(points, k, maxIter)
         #6
      M[i] ← intraClusterDistance(points, centroids,
clusters)       #7
    results[k] ← min(M)
                                              #8
  return results
                #9
```

#1 Method `kmeansElbow` takes a list of points, a list of values to test for `k` (aka `numCentroids`), the number of clustering it should create, `maxIter`, the maximum number of iterations for k-means, and how many runs of the algorithm are needed for each value of `k` tested. This function returns an associative array with the best value for the intra-cluster distance metric obtained for each number of centroids tested.
#2 Initialize the associative array with the results.
#3 Cycle through all the values for `k` to test.
#4 Initialize an array to keep track of the results (because we are running the algorithm several times for each value of `k`).
#5 Repeat `runsPerK` times.
#6 Run k-means with the given input and hyper-parameters.
#7 Compute the intra-cluster distance metric on the output of the algorithm.
#8 Choose the best result of the intra-cluster distance metric among all runs with the current value for the number of